FROM DIAGONAL PRODUCTS TO A CHANGE OF SIGN

Difference of two squares

Squares, cubes and a change of sign through two small determinants.

1 / START WITH A TWO-BY-TWO DETERMINANT

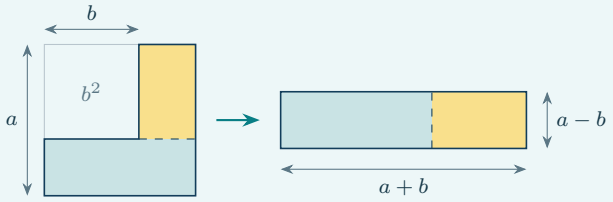
The difference of squares is a determinant

Subtract the products of the two diagonals:

$$D_2 = \begin{vmatrix} a & b \\ b & a \end{vmatrix} = a \cdot a - b \cdot b.$$

$$\boxed{D_2 = a^2 - b^2 = (a+b)(a-b)}.$$

The same area can be seen in two ways. For $a > b > 0$, remove the smaller square and rearrange the pieces.



$$8^2 - 5^2 = (8+5)(8-5) = 13 \times 3 = 39.$$

2 / MOVE EACH ROW ROUND ONE PLACE

And for a 3×3 matrix?

$$D_3 = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}.$$

The six determinant terms give

$$D_3 = a^3 + b^3 + c^3 - abc - abc - abc.$$

$$\boxed{D_3 = a^3 + b^3 + c^3 - 3abc}.$$

Set $c = 0$: this becomes $a^3 + b^3$. Now replace b by $-b$ to obtain the difference:

$$\boxed{a^3 - b^3 = (a-b)(a^2 + ab + b^2)}.$$

The middle ab is still needed:

$$(a-b)(a^2+b^2) \neq a^3 - b^3 \quad \text{in general.}$$

3 / LET THE SECOND TERM BECOME IMAGINARY

A sum of squares can factorise too

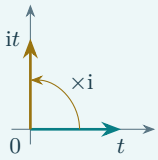
The imaginary unit is defined by $\mathrm{i}^2 = -1$. Hence

$$(\mathrm{i}b)^2 = \mathrm{i}^2 b^2 = -b^2.$$

Replace b by $\mathrm{i}b$ in the familiar identity:

$$a^2 + b^2 = a^2 - (\mathrm{i}b)^2$$
$$= \boxed{(a + \mathrm{i}b)(a - \mathrm{i}b)}.$$

$$\begin{vmatrix} a & \mathrm{i}b \\ \mathrm{i}b & a \end{vmatrix} = a^2 + b^2.$$



Multiplying by i makes a quarter-turn in the complex plane.

Squaring gives the sign change:

$$(\mathrm{i}t)^2 = -t^2.$$

4 / A SMALL GLIMPSE OF PHYSICS

From t to $\mathrm{i}t$: Wick rotation

Try $t \mapsto \mathrm{i}t$ in a difference of squares:

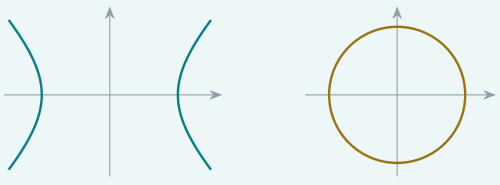
$$x^2 - t^2 \longrightarrow x^2 - (\mathrm{i}t)^2 = \boxed{x^2 + t^2}.$$

In physics, space and time can appear with opposite signs. Writing $t = \mathrm{i}\tau$, with τ real, gives

$$x^2 - c^2 t^2 \longrightarrow x^2 + c^2 \tau^2.$$

Here c is the speed of light. This is the sign change behind **Wick rotation**, which relates spacetime formulas to Euclidean geometry.

For real coordinates, compare these two curves:



$x^2 - y^2 = 1$
hyperbola

$x^2 + y^2 = 1$
circle

The same matrix pattern, with one change of sign

$$\begin{vmatrix} a & b \\ b & a \end{vmatrix} = a^2 - b^2 \qquad \begin{vmatrix} a & \mathrm{i}b \\ \mathrm{i}b & a \end{vmatrix} = a^2 + b^2.$$

Physics link. An imaginary-time substitution is a mathematical change of coordinate used in calculations. Further reading: J. Penedones and S. Reymond, *Quantum Physics IV*, EPFL, Lecture 5.

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